

STAT842 Final Project: Effects of Color and Labeling on Line Graph Reading Accuracy

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1 Introduction

Line graphs are very commonly used plots, especially in showing the change of numerical values over time. Small design choices, such as using one color or many colors and placing labels inside the graph or in a legend, may change how accurately people read values from a graph. Many design guidelines are based on experience or intuition, but there is still room to test how these choices affect accuracy in a controlled setting.

In our project, we studied a 2×2 factorial experiment. The first factor was **Color**: a single-color line with different lightness levels versus a line graph where each line has its own color. The second factor was **Label type**: a separate legend versus inline labels placed near the lines. We combined these two factors to form four conditions: single + legend, single + inline, color + legend, and color + inline. Through experiments, we evaluated these factors' impact on the accuracy of reading data from a line graph.

Intuitively, we believed that using different colors and inline labels both improve accuracy because these designs make the differences between lines in the image more obvious, reducing identification errors and increasing accuracy. Some rigorous statistical methods will be used to verify/refute these intuitions.

2 Design

2.1 Survey Design

An online survey (https://uwaterloo.ca1.qualtrics.com/jfe/form/SV_8erzqICX0Ge4EF8) was distributed to students of STAT842 randomly for data collection. The survey participants were expected to see a 'Participant Instruction' on the first glance, which explained the project background and educated people to complete the survey correctly.

The graphs were based on Official Canadian Climate Normals (1991–2020), Daily Average ($^{\circ}\text{C}$), from Environment and Climate Change Canada (ECCC) [1]. Each graph showed monthly average temperatures for a subset of cities across the twelve months. For each design condition, students saw one line graph and answered three questions about it: In addition to the density plots, boxplots, and the five-number summary of `|temp_err|`, we examined three types of temperature-related tasks. First, **Q1** is a single numeric value query, where participants answer questions such as "What is the temperature of city X in month Y?" by providing an open numeric response in $^{\circ}\text{C}$. Second, **Q2** is a numeric difference task, in which participants estimate a temperature change, for example "By how many degrees does city X warm from March to July?", again reported as a difference in $^{\circ}\text{C}$. Finally, **Q3** is an average-level comparison task, where participants answer questions such as "Which city

has the highest mean temperature over the year?” using a multiple-choice response between two cities.

Based on the chart, please answer the following questions carefully.

★ What is the temperature (°C) of Toronto in September?

Please round to one decimal place.

★ By how many degrees (°C) does Montreal warm up from March to July?
(July temp - March temp)

Please round to one decimal place.

★ Which city has higher **annual mean** temperature?

a. Vancouver

b. Montreal

Please input the option (a/b).

Figure 1: Participants answered (i) a single-point lookup, (ii) a difference between two months, and (iii) a categorical comparison about which city has the higher annual mean.

These three questions were designed to represent different types of tasks: reading a single point from the graph, comparing two points and computing a difference, and identifying the overall average.

In addition to these three accuracy questions, we also collected confidence ratings for each graph. After answering the three questions for a given graph, participants were asked: “Based on the questions above, how confident do you feel about reading data from this graph?”, and they responded on a 0–10 scale (0 = not confident at all, 10 = very confident). Each participant gave one confidence rating for each of the four design conditions, so the confidence data follow the same 2×2 within-subject structure as the accuracy data.

2.2 Experiment Design

We used a 2×2 within-subjects design. Each participant saw all four design conditions: single-color line with legend, single-color line with inline labels, multi-color lines with legend, multi-color lines with inline labels.

For each condition we created three parallel question sets (blocks). Within a block, the structure of the three questions remained the same, but the specific cities and months changed. We tried to keep the difficulty of these sets similar through careful design. Each participant was randomly assigned to one of the three sets, so each student answered one set per condition and four sets in total.

2.3 Randomization

To mitigate potential confounding, we implemented three layers of design control and randomization.

First, the survey link was distributed to all students in the STAT 842 course, and participation was voluntary. Although this is a convenience sample rather than true random sampling from a broader population, restricting participants to the same course helps reduce large between-subject differences in statistical background (e.g., undergraduate vs. graduate status).

Second, each participant was required to complete all four conditions, but the displaying order of the four conditions was individually randomized. This within-subject randomization reduces systematic learning or fatigue effects that could otherwise be confounded with the visualization condition.

Third, for each condition we created three parallel question blocks with identical structure but different cities, numerical values in the line graphs and corresponding questions. Participants were randomly assigned to one block per condition, and we manually designed the blocks to be similar in difficulty. This prevents participants from simply recalling previous answers instead of reading each new graph and helps to control for memory and practice effects. In addition, the color schemes also varied across these parallel blocks, which ensures that we are defining the levels of the color factor more generally, rather than merely comparing two specific color schemes.

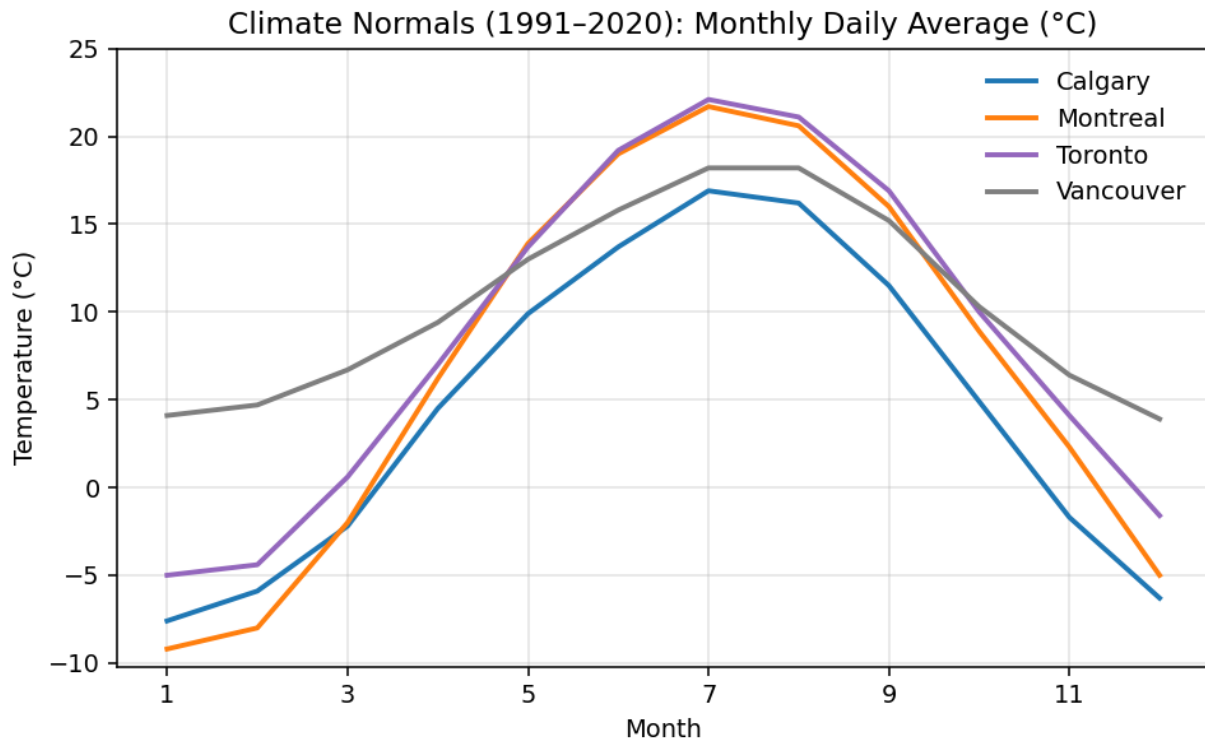


Figure 2: Monthly daily average temperatures (1991–2020) are plotted across months for four Canadian cities; graphs of this form were used in the survey tasks.

3 Results

There were 93 complete responses without any missing value collected from students. All analysis were carried out in Python.

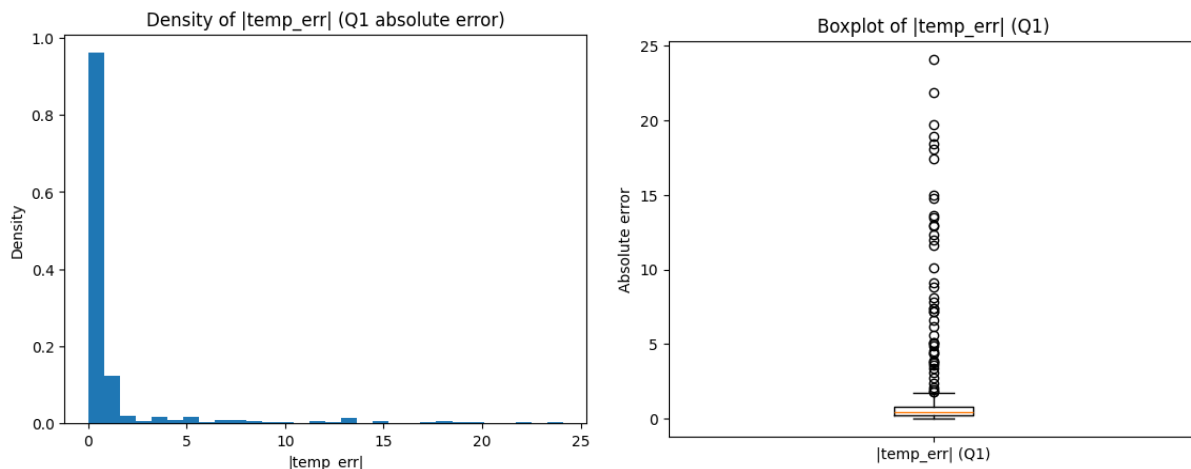
For data processing, we converted the data to a long format with one row per participant $\times$ condition. For each row we stored: subject (participant anonymous ID), condition (one of the four

combinations), color_scheme (single vs color), label_type (legend vs inline), temp_err (response – true value for Q1), warmup_err (response – true value for Q2), annual01 (1 if answer to Q3 was correct, 0 otherwise), confidence (rate in confidence question).

We then computed absolute errors: abs_temp_err ($|\text{temp_err}|$) and abs_warmup_err ($|\text{warmup_err}|$).

3.1 EDA

Firstly, we conducted exploratory data analysis. For Q1, Figure 3 summarizes the distribution of $|\text{temp_err}|$ using both a density plot and a boxplot. Panel (a) shows that most absolute errors were below a few degrees, with a sharp peak near zero and a modest right tail. Panel (b) confirms this pattern, with a low median and a tight interquartile range, and only a small number of higher outliers. This indicates that the single-value temperature question was easy for most students to answer accurately. To summarize these patterns numerically, Table 1 reports the five-number summary of $|\text{temp_err}|$ by condition.



(a) Most errors cluster near zero with a modest right tail, suggesting Q1 was generally easy. (b) The median and IQR are small, with a limited number of high outliers.

Figure 3: Distribution and boxplot of $|\text{temp_err}|$ (Q1).

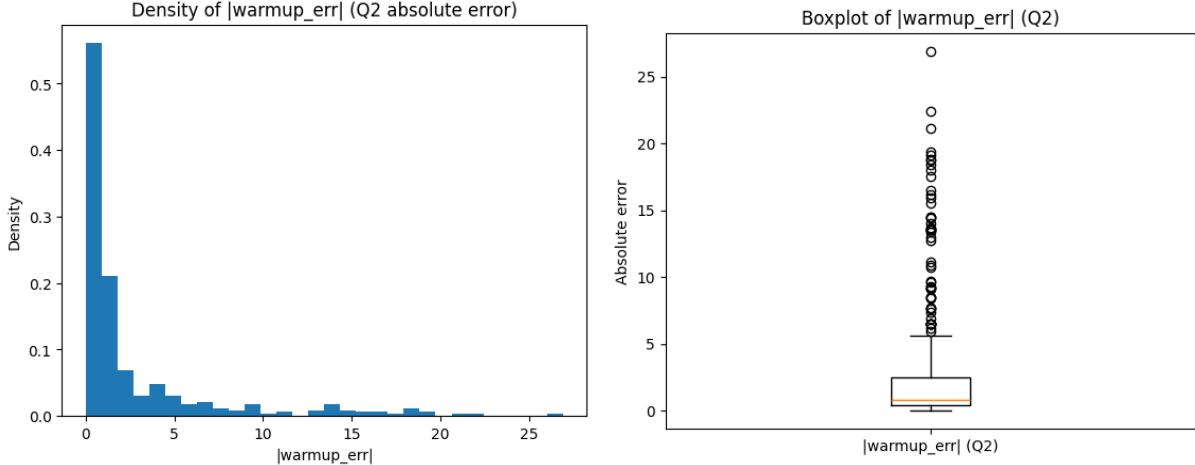
Table 1: Five-number summaries of $|\text{temp_err}|$ (Q1). In all four design conditions, the minimum, median, and upper quartiles are close to zero, indicating small errors and high accuracy, with only a few larger responses driving the maximum values. This pattern agrees with the Q1 density plot and boxplot.

Condition	Minimum	First Quartile (q1)	Median (q2)	Third Quartile (q3)	Maximum
Single + Legend	0.0	0.2	0.3	0.7	19.7
Color + Legend	0.0	0.1	0.4	0.9	24.1
Single + Inline	0.0	0.2	0.3	0.8	18.9
Color + Inline	0.0	0.2	0.4	0.9	18.4
Overall	0.0	0.2	0.4	0.8	24.1

For Q2, we first inspected the distribution of $|\text{warmup_err}|$ using an overall density plot. The distribution showed a very long right tail with two extreme values. After checking the corresponding responses, we found that these two absolute errors (about 85°C and 199°C) were almost certainly

data-entry errors, likely caused by a missing decimal point, because such changes are far outside any realistic temperature range. We therefore removed these two rows from the Q2 dataset and conducted analyses on the cleaned data.

The density plot of $|\text{warmup_err}|$ shows a clear peak near zero and a strongly right-skewed shape, which means that most responses involve only small errors, with a minority of cases producing larger warm-up differences and a long right tail. The overall boxplot tells a consistent story, with a low central tendency and a relatively narrow core spread, plus a longer upper whisker and a few genuinely large errors. Table 2 provides the corresponding five-number summary of $|\text{warmup_err}|$ by condition.



(a) Errors are right-skewed with a long tail, reflecting occasional large mistakes. (b) Most errors are small, but a few very large values appear as extreme outliers.

Figure 4: Distribution and boxplot of $|\text{warmup_err}|$ (Q2).

Table 2: Five-number summaries of $|\text{warmup_err}|$ (Q2). Compared with Q1, the medians and upper quartiles are larger and some maxima are substantially higher, indicating greater errors and a longer right tail for the warm-up task. This pattern is consistent with the Q2 density plot and boxplot.

Condition	Minimum	First Quartile (q1)	Median (q2)	Third Quartile (q3)	Maximum
Single + Legend	0.0	0.3	0.6	1.9	18.8
Color + Legend	0.0	0.3	1.2	3.5	22.4
Single + Inline	0.0	0.5	1.1	4.0	26.9
Color + Inline	0.0	0.4	0.5	2.5	19.1
Overall	0.0	0.4	0.8	2.5	26.9

For Q3, we looked at the proportion of correct answers in each condition. The interaction plot of mean correctness by Color and Label showed that accuracy was high in all cells, and the line for inline labels lay slightly above the line for legends across both color schemes. The bar plot by condition showed a similar pattern: both inline conditions had slightly higher correctness than their legend counterparts, but the differences were small.

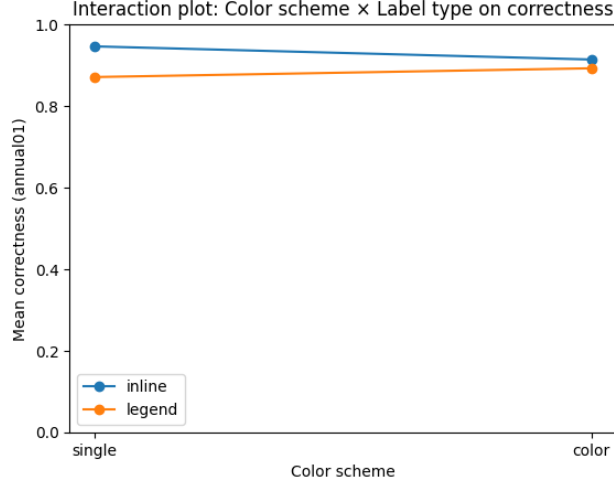


Figure 5: Mean correctness is high across conditions; inline labeling appears slightly higher than legend across both color schemes.

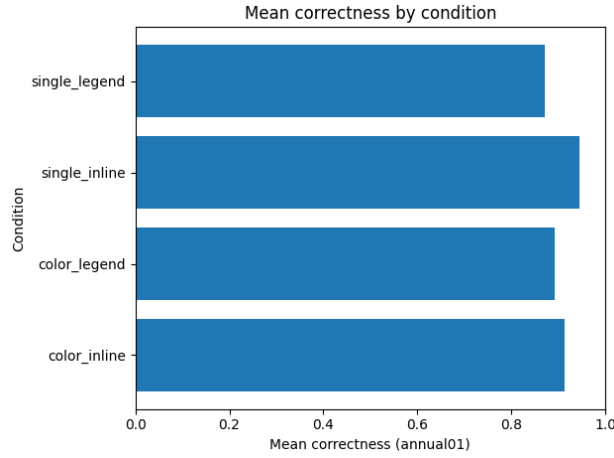


Figure 6: Inline conditions show slightly higher correctness than their legend counterparts, though differences are small.

Besides, because the distributions of absolute errors were right-skewed, especially for Q2, we used log-transformed absolute errors as our main accuracy measures:

$$\text{Q1 outcome: } y_{ic}^{(1)} = \log(|\text{temp_err}_{ic}| + 1), \quad \text{Q2 outcome: } y_{ic}^{(2)} = \log(|\text{warmup_err}_{ic}| + 1).$$

These transformed measures reduce the impact of very large errors and give a more symmetric outcome for linear modelling.

3.2 Model Fitting

Each participant answered all four conditions, so their responses are not independent. To respect this paired structure, we used models with participant-level random effects, instead of simple independent-samples tests.

For Q1 and Q2 we analyzed the log-transformed absolute errors with linear mixed-effects models. In both models we included Color (single vs color), Label (legend vs inline), and their interaction as fixed effects, and we added a random intercept for participant to capture baseline differences in ability.

For Q3 the outcome is binary (correct vs incorrect). A linear model would not be appropriate. We therefore used a logistic GEE model with a logit link. We used the same fixed-effect structure (Color, Label, and their interaction) and treated participant ID as the clustering variable with an exchangeable working correlation.

We also conducted model diagnostics for the mixed models in Q1 and Q2. Residual plots showed some right skew and some increase in variance with fitted values, especially for Q1, so the normal and constant-variance assumptions are only approximately met. However, there were no extreme residual outliers after the log transformation, and the main patterns of significance were stable across reasonable alternative specifications. Considering the limited sample size, we believed that a linear mixed-effects model remains the most appropriate. For Q3, because the predictors are categorical and each condition contained both correct and incorrect responses, the key logistic assumptions of linearity in the logit and no complete separation appear reasonable.

3.2.1 Q1: Mixed model for single-value reading

For Q1, we defined the accuracy as $y_{ic}^{(1)}$, where i indexes participants and c indexes the four graph conditions. Then we fitted the following linear mixed-effects model with a random intercept for participant:

$$y_{ic}^{(1)} = \beta_0 + \beta_1 \text{Color}_{ic} + \beta_2 \text{Label}_{ic} + \beta_3 (\text{Color}_{ic} \times \text{Label}_{ic}) + b_i + \varepsilon_{ic},$$

where Color = 0 for single color and 1 for multiple color; Label = 0 for legend and 1 for inline; β_0 is the mean log absolute error in the baseline condition; β_1 is the main effect of Color, β_2 is the main effect of Label, β_3 is the interaction; b_i is the random intercept for participant i ; and ε_{ic} is the residual error term.

According to the analysis results of Q1, the intercept of the mixed model was about 0.55 ($p < 0.001$), which corresponds to the single-color + legend condition. When we back-transform this value, it implies a typical absolute error of about 0.8°C . The Color main effect was small (0.06, $p = 0.42$), and the Label main effect was also small (-0.07 , $p = 0.41$). The interaction was close to zero (-0.09 , $p = 0.42$). None of these effects were significant. The variance of the participant random intercepts was about 0.16, compared to a residual variance of about 0.29, which means that some students were consistently more accurate than others. This supports the use of a mixed-effects model rather than treating all observations as independent.

3.2.2 Q2: Mixed model for difference reading

For Q2, we had the same model structure as Q1. The model outputs showed that the intercept (0.80, $p < 0.001$) again corresponds to the single-color + legend condition and translates to an absolute error of about 1.2°C when estimating the temperature difference between March and July. Within legend-labelled graphs, the Color main effect (0.12, $p = 0.20$) was not significant, which suggests that switching from a single-color line to a multi-color line, with legend, did not reliably change accuracy. Within single-color graphs, the Label main effect (0.25, $p = 0.006$) was positive and significant. It indicates that when color is single, switching from a legend to inline labels increased error. Roughly, participants made 25% more error with inline labels than with a legend in single-color line graphs.

The interaction was -0.37 ($p = 0.004$), so the effect of Label clearly depended on the color scheme. Furthermore, Table 3 reports the predicted typical absolute errors on the original scale.

Table 3: Absolute Warm-up Error ($^{\circ}\text{C}$) by Condition. Predicted typical absolute errors on the original scale for Q2.

Condition	Absolute error ($^{\circ}\text{C}$)
Single + Legend	1.2
Color + Legend	1.5
Single + Inline	1.9
Color + Inline	1.2

Furthermore, Table 3 shows that inline labels are harmful in single-color graphs, but in multi-color graphs they can reduce the extra visual complexity and bring performance back to the baseline level. The random intercept variance for Q2 was also quite large, again indicating differences in accuracy between participants.

3.2.3 Q3: Logistic GEE for correctness

For Q3 the logistic GEE model used correctness (1 vs 0) as the outcome:

$$\text{logit}(\Pr(\text{annual01}_{ic} = 1)) = \gamma_0 + \gamma_1 \text{Color}_{ic} + \gamma_2 \text{Label}_{ic} + \gamma_3 (\text{Color}_{ic} \times \text{Label}_{ic}).$$

In the baseline condition (single-color + legend), the intercept (1.90, $p < 0.001$) corresponds to a predicted correctness of about 87%. Thus, even the basic design already yielded high accuracy on this categorical task. The Color main effect (0.22, $p = 0.63$) was not significant, so we found no clear evidence that multi-color lines were better or worse than single-color lines for Q3. The Label main effect (1.20, $p = 0.039$) was positive and significant. For single-color graphs, this means that using inline labels increased the odds of a correct answer by about a factor of three. The predicted correctness rose from about 87% with a legend to about 95% with inline labels in the single-color condition. The interaction (-0.95 , $p = 0.18$) was not significant. Using the fitted model, the predicted correctness for the four conditions was roughly: 87% for single + legend, 89% for color + legend, 96% for single + inline, 91% for color + inline. In summary, Q3 accuracy was high in all four conditions, with some evidence that inline labels can further improve performance, especially in single-color graphs.

3.2.4 Confidence ratings

Finally, we analyzed the 0–10 confidence ratings that participants gave after each graph. Mean confidence was around 4.8–4.9 for the two single-color conditions (4.80 for single + legend, 4.94 for single + inline), and around 6.7–6.8 for the two multi-color conditions (6.74 for color + legend, 6.77 for color + inline). A random-intercept linear mixed model with Color, Label, and their interaction as fixed effects showed a large and highly significant main effect of Color (multi vs single; estimate ≈ 1.94 , SE ≈ 0.22 , $p < 0.001$), but no significant main effect of Label (inline vs legend; estimate ≈ 0.13 , $p \approx 0.55$) and no significant interaction (estimate ≈ -0.10 , $p \approx 0.75$). Thus, multi-color graphs increased subjective confidence by almost two points on the 0–10 scale, whereas label placement had little impact on confidence and did not change the impact of color.

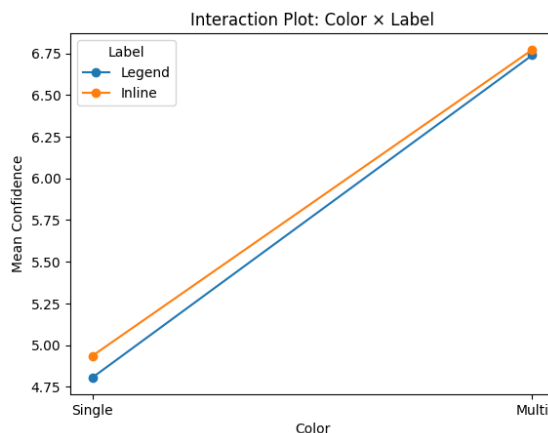


Figure 7: Mean confidence increases sharply from single-color to multi-color graphs. Inline labels show a small additional increase over legends, and the two lines are nearly parallel, suggesting little to no interaction between Color and Label for confidence.

4 Conclusion

This project was conducted to evaluate how two design factors, Color and Label, influence line graph accuracy. In Q1, a simple single value reading task, accuracy was high in all four conditions, and the mixed model found no significant effects of Color, Label, or their interaction, suggesting that these design variants have little impact on how accurately students read a line graph in this setting. Thus our initial intuition about “more colorful and more labeled is always better” does not hold in this case.

In Q2, which asked for a temperature difference between two months, design choices had a clear impact on accuracy. Inline labels led to larger errors when they were combined with a single-color line, but they were helpful when combined with multi-color lines and brought performance back to the level of the single + legend baseline. Based on this result, when readers need to judge changes or differences, we recommend avoiding the single-color + inline combination and instead using single-color + legend or multi-color + inline in terms of data reading accuracy.

For the task in Q3, the logistic GEE model showed no clear effect of Color and only a small advantage for inline labels, especially in single-color graphs, so our initial expectation of large accuracy gains from multiple colors and inline labels was not strongly supported.

These findings show that Color and Label can meaningfully affect line graph accuracy in some situations, but even in the worst-performing condition, typical errors remained on the order of only a few degrees Celsius, so the design effects are meaningful but not huge. The confidence data point in a similar direction: multi-color graphs make participants feel more confident reading values from a graph, whereas label placement (inline vs legend) has little effect on confidence and does not change the impact of color. In other words, our initial intuition that “more color” helps is clearly reflected in confidence, but only partly reflected in objective accuracy.

This study also had an important limitation. Within each design condition we used three different but structurally similar question sets, and each participant saw only one set per condition. We tried to keep the difficulty of these sets similar, but they may not be perfectly matched. Some differences between conditions may therefore reflect differences in item difficulty rather than pure design effects.

References

- [1] Environment and Climate Change Canada (ECCC). *Canadian Climate Normals 1991–2020: Daily Average (°C)*. Government of Canada. Available at https://climate.weather.gc.ca/climate_normals/index_e.html. Accessed December 11, 2025.